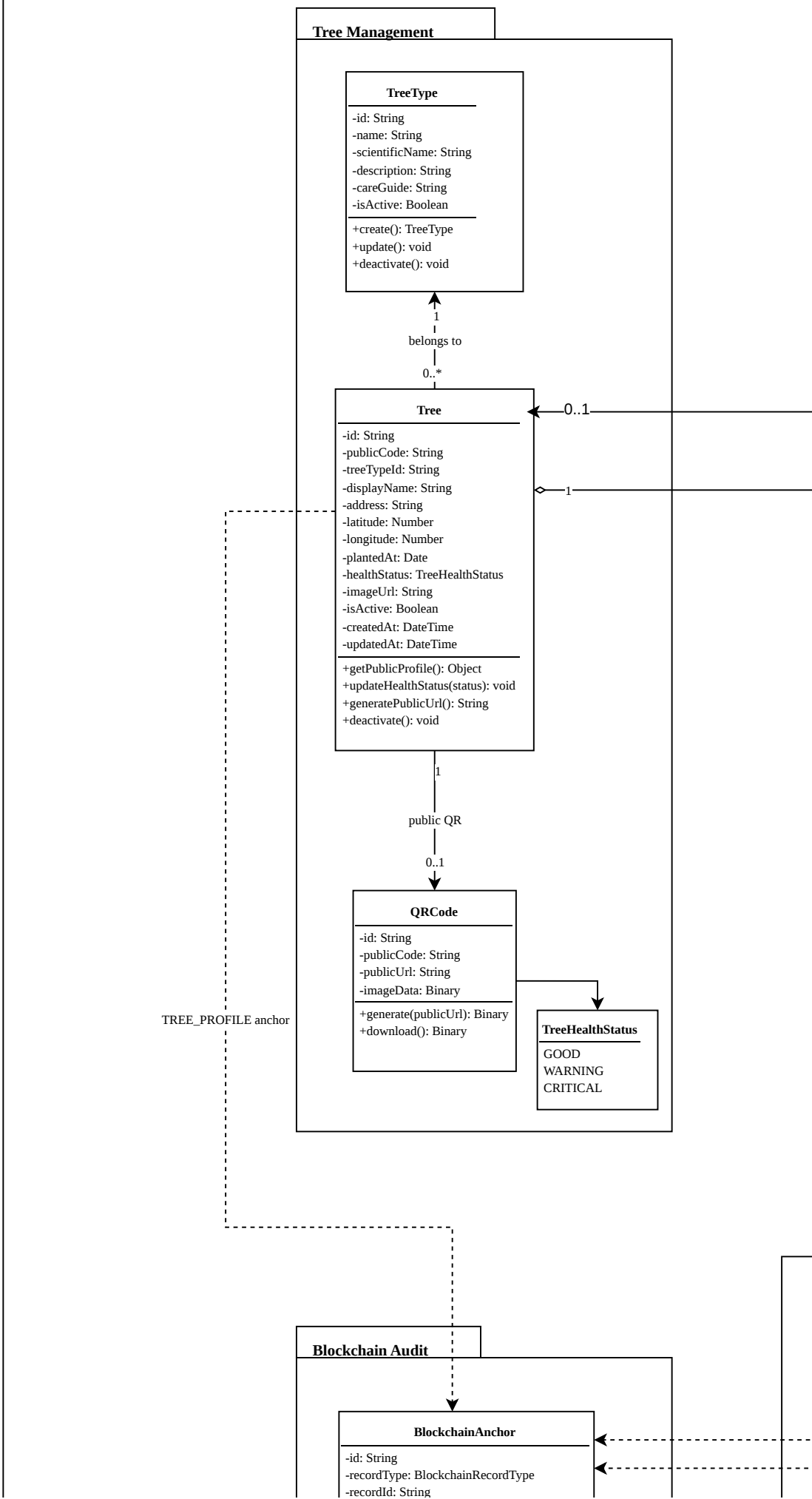
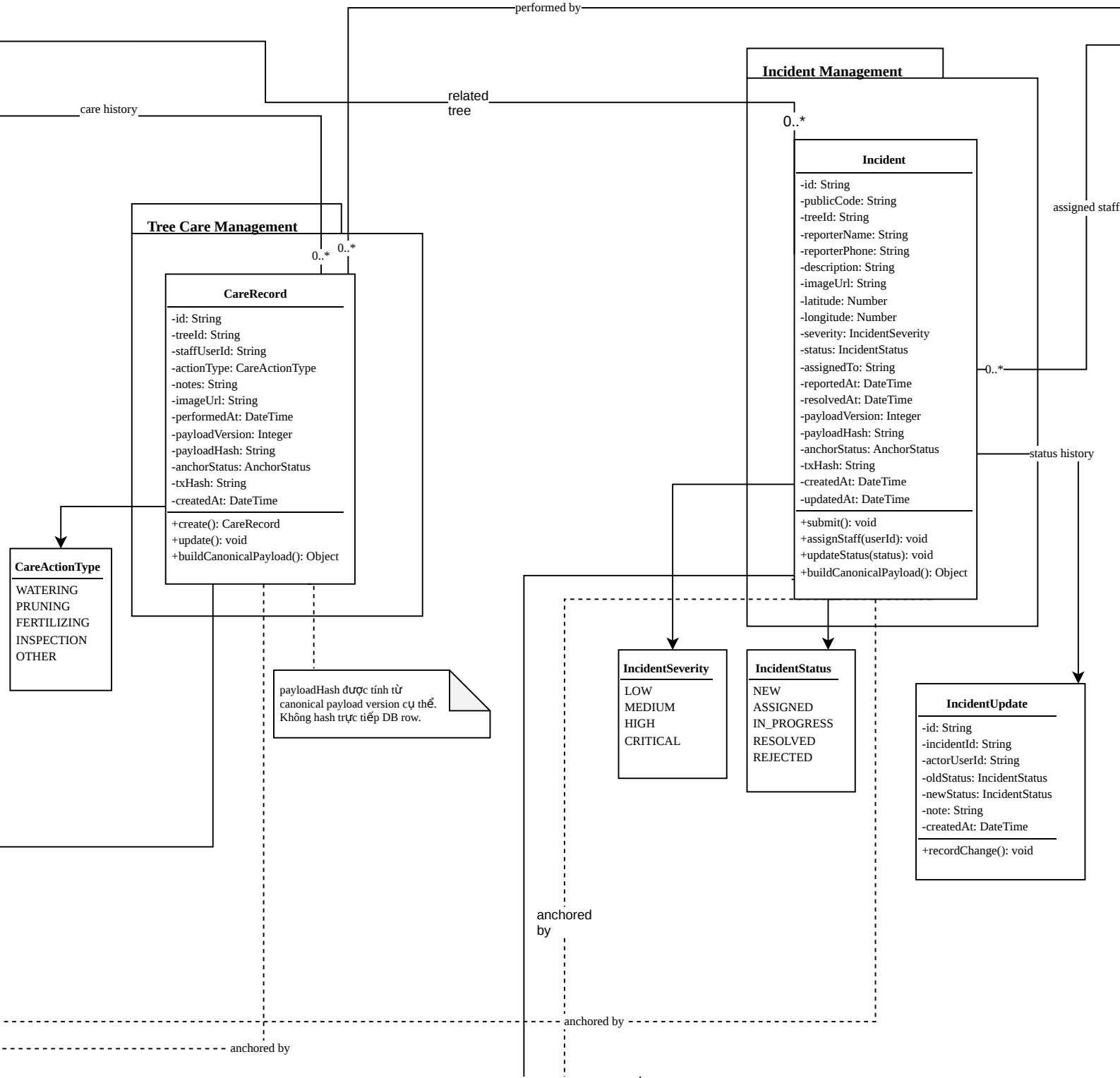
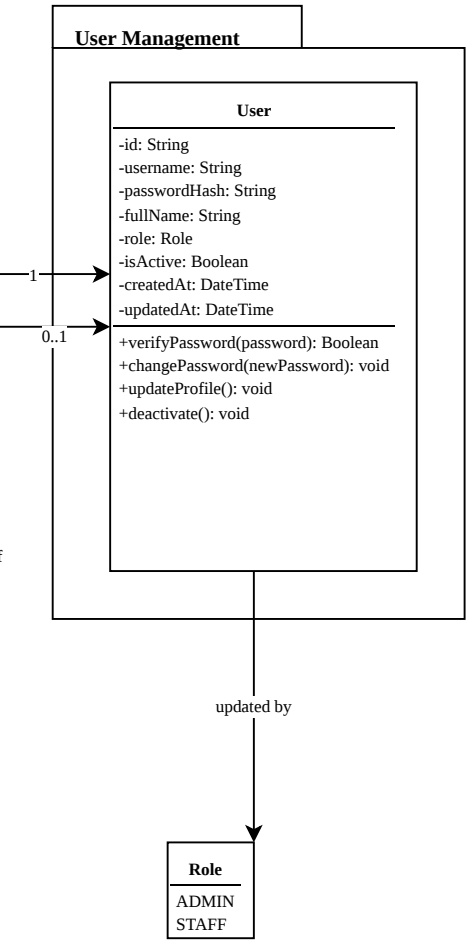
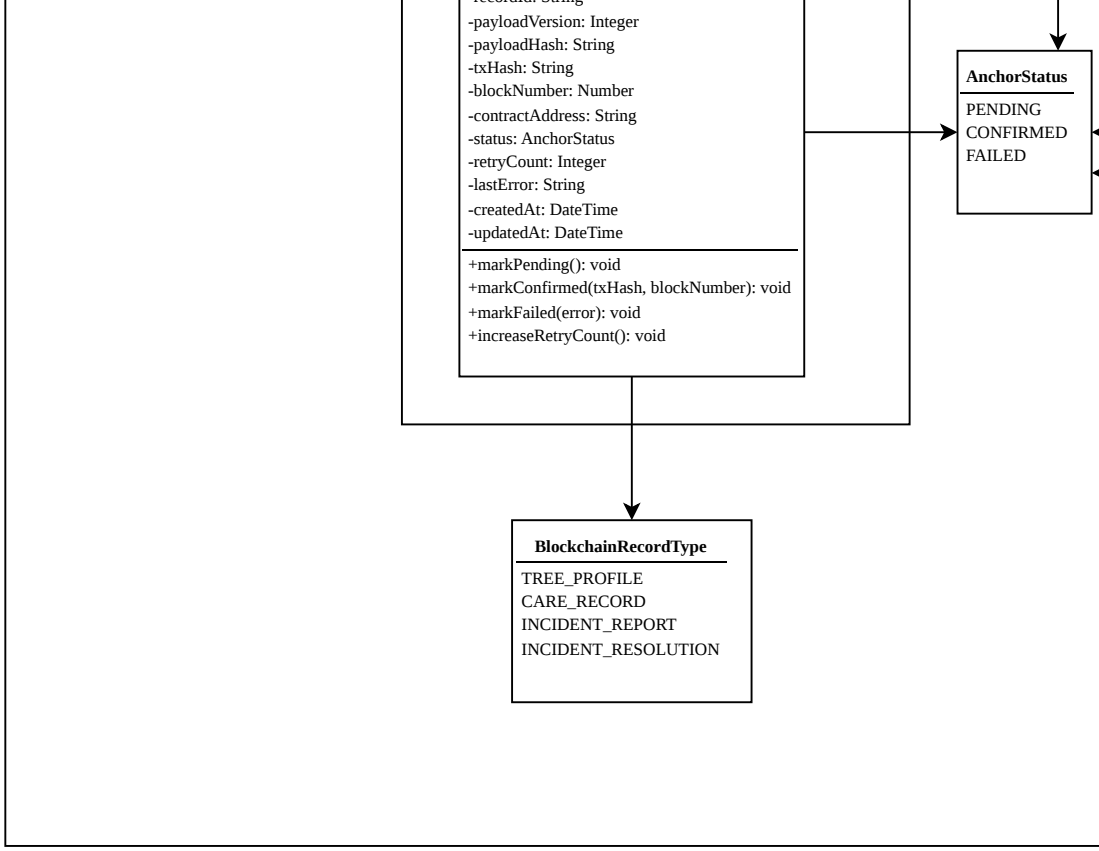

Tree Management
TreeType
-id: String
-name: String
-scientificName: String
-description: String
-careGuide: String
-isActive: Boolean
+create(): TreeType
+update(): void
+deactivate(): void
1
belongs to
0..*
Tree
-id: String
-publicCode: String
-treeTypeId: String
-displayName: String
-address: String
-latitude: Number
-longitude: Number
-plantedAt: Date
-healthStatus: TreeHealthStatus
-imageUrl: String
-isActive: Boolean
-createdAt: DateTime
-updatedAt: DateTime
+getPublicProfile(): Object
+updateHealthStatus(status): void
+generatePublicUrl(): String
+deactivate(): void
0..1
1
1
public QR
0..1
QRCode
-id: String
-publicCode: String
-publicUrl: String
-imageData: Binary
+generate(publicUrl): Binary
+download(): Binary
TreeHealthStatus
GOOD
WARNING
CRITICAL
TREE_PROFILE anchor
Blockchain Audit
BlockchainAnchor
-id: String
-recordType: BlockchainRecordType
-recordId: String

URBANTREE - DOMAIN CLASS DIAGRAM




User Management
User
-id: String
-username: String
-passwordHash: String
-fullName: String
-role: Role
-isActive: Boolean
-createdAt: DateTime
-updatedAt: DateTime
+verifyPassword(password): Boolean
+changePassword(newPassword): void
+updateProfile(): void
+deactivate(): void
1
0..1
updated by
Role
ADMIN
STAFF


-payloadVersion: Integer
-payloadHash: String
-txHash: String
-blockNumber: Number
-contractAddress: String
-status: AnchorStatus
-retryCount: Integer
-lastError: String
-createdAt: DateTime
-updatedAt: DateTime
+markPending(): void
+markConfirmed(txHash, blockNumber): void
+markFailed(error): void
+increaseRetryCount(): void
AnchorStatus
PENDING
CONFIRMED
FAILED
BlockchainRecordType
TREE_PROFILE
CARE_RECORD
INCIDENT_REPORT
INCIDENT_RESOLUTION

